Dimensional Repair and Refined Framework for Entanglement-Entropy Sourcing of Gravity

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Abstract

This white paper repairs a critical dimensional inconsistency in earlier drafts of the entanglement–gravity framework and presents a refined, self-consistent phenomenological model. Earlier work identified a covariant structure and an experimental program, but treated the entanglement entropy density $S_{\text{ent}}(x)$ as entering the stress–energy tensor without the physical length or energy scale required in SI units. Here we introduce the minimal additional scale needed to make the entanglement contribution a bona fide stress–energy tensor, redefine the effective action accordingly, and re-express the observable combination $\mathcal{B}$ and the fifth-force equation with correct dimensions. The OP3 screening program and NISQ experimental outlook are preserved, but are now placed on a dimensionally sound foundation.

1 Purpose and Scope

This document is a dimensional-consistency revision of the entanglement–gravity program. Its goal is not to select a unique microscopic origin for the coupling, but to isolate the minimal scaling assumptions required for a well-posed phenomenological theory.

Earlier drafts used a coupling prefactor proportional to $c^4/(8\pi G k_B \ln 2)$ multiplying an entanglement entropy density measured in bits per unit volume. In SI units, that expression does not yield an energy density, so the resulting stress–energy tensor and derived accelerations were not dimensionally valid. The present revision fixes that problem by introducing a reference energy scale E_* and rebuilding the source term around it.

2 Units and the Original Problem

The Einstein equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1}$$

require the stress–energy tensor to have units of energy density,

$$[T_{\mu\nu}] = \mathrm{J\,m^{-3}} = \mathrm{kg\,m^{-1}\,s^{-2}}. \tag{2}$$

The original framework defined the coarse-grained entanglement entropy density as

$$[S_{\text{ent}}] = \mathrm{bit\,m^{-3}}, \tag{3}$$

where "bit" is dimensionless for unit counting. The problematic source coefficient was written as

$$\alpha = \tilde{\kappa}\,\frac{c^4}{8\pi G k_B \ln 2}. \tag{4}$$

In SI units, this gives

$$[\alpha] = \mathrm{J\,m^{-1}\,K^{-1}}, \tag{5}$$

so that

$$[\alpha S_{\mathrm{ent}}] = \mathrm{J\,m^{-4}\,K^{-1}}, \tag{6}$$

which is not an energy density. A missing physical scale is therefore required to convert entropy density into energy density.

3 Minimal Dimensional Fix

3.1 Reference energy scale

We introduce a reference energy scale E_* with units of energy per bit and define the entanglement energy density by

$$\rho_{\mathrm{ent}}(x) = E_*\, S_{\mathrm{ent}}(x). \tag{7}$$

Then

$$[\rho_{\mathrm{ent}}] = [E_*][S_{\mathrm{ent}}] = \mathrm{J\,m^{-3}}, \tag{8}$$

as required.

An equivalent parameterization is

$$\rho_{\mathrm{ent}}(x) = \lambda_* k_B T_* s_{\mathrm{ent}}(x), \tag{9}$$

where s_{ent} is entropy density in thermodynamic units, T_* is an effective temperature scale, and λ_* is dimensionless. In this white paper we use the simpler single-scale notation E_*.

3.2 Revised source term

The entanglement contribution to the stress–energy tensor is rewritten as

$$T^{\mathrm{ent}}_{\mu\nu}(x) = \tilde{\kappa}\, \rho_{\mathrm{ent}}(x) g_{\mu\nu} + (\text{gradient terms}), \tag{10}$$

with

$$\rho_{\mathrm{ent}}(x) = E_* S_{\mathrm{ent}}(x). \tag{11}$$

Thus all previous appearances of αS_{ent} are replaced by $\tilde{\kappa} E_* S_{\mathrm{ent}}$.

If desired, one may later relate E_* to a microscopic scale such as

$$E_* = \eta k_B T_*, \tag{12}$$

with η dimensionless and T_* an effective Unruh or horizon temperature. The present framework leaves the origin of E_* open.

4 Revised Effective Action and Fifth-Force Equation

4.1 External-source action

Let

$$\phi(x) \equiv S_{\mathrm{ent}}(x). \tag{13}$$

We retain the external-source interpretation: ϕ is prescribed by the device quantum state and is not varied as an independent field. The entanglement-sector Lagrangian density is written as

$$\mathcal{L}_{\text{ent}} = -\frac{\beta}{2} g^{\mu\nu} (\nabla_\mu \phi)(\nabla_\nu \phi) - \lambda \phi, \tag{14}$$

with

$$\lambda = \tilde{\kappa} E_*. \tag{15}$$

The full action is therefore

$$S[g;\phi] = \int d^4x \sqrt{-g} \left[\frac{c^3}{16\pi G} R - \frac{\beta}{2} g^{\mu\nu} (\nabla_\mu \phi)(\nabla_\nu \phi) - \tilde{\kappa} E_* \phi \right]. \tag{16}$$

Varying with respect to $g_{\mu\nu}$ yields

$$T^{\text{ent}}_{\mu\nu} = \beta \left(\nabla_\mu \phi \, \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} (\nabla \phi)^2 \right) + \tilde{\kappa} E_* \phi \, g_{\mu\nu}. \tag{17}$$

The dimensions are correct provided β carries the dimensions needed to make the gradient term an energy density. In slowly varying configurations, the gradient term is subleading.

4.2 Matter nonconservation

Total conservation requires

$$\nabla^\mu T^{\text{total}}_{\mu\nu} = 0, \tag{18}$$

so that

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu}. \tag{19}$$

Substituting the revised entanglement tensor gives

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\left(\tilde{\kappa} E_* + \beta \Box \phi \right) \nabla_\nu \phi + \beta \nabla_\nu \left(\frac{1}{2} (\nabla \phi)^2 \right). \tag{20}$$

In the quasi-static regime with modest gradients and $|\beta \Box \phi| \ll |\tilde{\kappa} E_*|$, this reduces to

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} \approx -\tilde{\kappa} E_* \, \nabla_\nu \phi(x) = -\tilde{\kappa} E_* \, \nabla_\nu S_{\text{ent}}(x). \tag{21}$$

This is the corrected fifth-force law. Its units are now those of force density, as required.

5 Newtonian Limit and Revised Observable

5.1 Modified Poisson equation

In the Newtonian limit,

$$\nabla^2 \Phi = 4\pi G \left(\rho_{\text{matter}} + \rho_{\text{ent}} \right), \tag{22}$$

with

$$\rho_{\text{ent}}(\mathbf{x}) = \tilde{\kappa} E_* S_{\text{ent}}(\mathbf{x}). \tag{23}$$

Under spherical symmetry,

$$\Delta a(R) = -\frac{\mathrm{d}\Phi_{\text{ent}}}{\mathrm{d}R} = \frac{G}{R^2} M_{\text{ent}}(< R), \tag{24}$$

where

$$M_{\text{ent}}(< R) = 4\pi \int_0^R \rho_{\text{ent}}(r) r^2 \, dr. \tag{25}$$

For bounded sources, $M_{\text{ent}}(< R)$ saturates outside the source and the correction scales as $1/R^2$.

5.2 Observable combination

In branch-comparison experiments, the useful quantity remains

$$\mathcal{B} \equiv |\tilde{\kappa}_{\rm eff}| \, \frac{\Delta M_{\rm ent}}{R_{\rm eff}^2}, \qquad \tilde{\kappa}_{\rm eff} = \tilde{\kappa} \, \alpha_{\rm screen}, \tag{26}$$

with

$$\Delta M_{\rm ent} = E_* \, \Delta S_{\rm ent}^{\rm tot}. \tag{27}$$

Therefore $\mathcal{B}$ has units of energy density, $\mathrm{J\,m^{-3}}$.

6 Implications for OP8 and OP10

6.1 Fifth-force bounds

The corrected acceleration scaling is

$$\Delta a(R) \sim \frac{G}{R^2} E_* \, \Delta S_{\rm ent}^{\rm tot}. \tag{28}$$

A null force bound $\Delta a_{\rm max}$ implies

$$E_* |\tilde{\kappa}| \lesssim \frac{\Delta a_{\rm max} R_{\rm eff}^2}{G \, \Delta S_{\rm ent}^{\rm tot}}. \tag{29}$$

This is the correct way to compare the theory with near-field force measurements or other Newtonian tests once a realistic entanglement inventory is specified.

6.2 RG interpretation

With E_* explicit, the natural renormalization-group ansatz becomes

$$\tilde{\kappa}_{\rm lab} E_{*,\rm lab} = \tilde{\kappa}_{\rm UV} E_{*,\rm UV} \left(\frac{E_{\rm lab}}{E_P} \right)^{\gamma}, \tag{30}$$

where E_P is the Planck energy and γ is an effective scaling exponent. Experiments therefore constrain the infrared product $\tilde{\kappa} E_*$ rather than $\tilde{\kappa}$ alone.

7 Programmatic Consequences

The dimensional repair does not alter the operational definition of $S_{\rm ent}(x)$, the Lindblad screening program OP3, or the computation of $\alpha_{\rm screen}(N, \tau)$. Those quantities remain meaningful exactly as before. What changes is only the mapping from entanglement content to physical source strength.

The attached analysis also argues that if one sets E_* near the Planck energy, a Ramsey-redshift experiment becomes far less promising than near-field force sensing, because the clock phase shift remains tiny while the local acceleration can be much larger. This suggests that optomechanical or levitated-mass force sensing may be the better experimental direction once the dimensional repair is accepted.

8 Conclusions

The entanglement–gravity framework can be made dimensionally consistent by introducing a reference energy scale E_* that converts entanglement entropy density into an energy density. With this repair, the external-source effective action, the entanglement stress–energy tensor, the matter nonconservation law, and the experimental observable $\mathcal{B}$ all acquire correct SI dimensions.

The open problems are now cleaner. Theory must constrain or derive E_* and its relation to ultraviolet physics; numerics must determine α_{screen} for realistic devices; and experiments must decide whether branch-comparison clocks or near-field force sensors provide the best route to bounding $\tilde{\kappa}E_*$. The key point is that the framework is no longer blocked by a units-level inconsistency and can now be developed as a well-posed phenomenological theory.